



MANIPAL ACADEMY OF HIGHER EDUCATION

FORMAL LANGUAGES AND AUTOMATA THEORY [CSE 2221]

Marks: 30

Duration: 90 mins.

MCQ

Answer all the questions.

Section Duration: 20 mins

- 1) Which of the following is true about the language accepted by an NFA and its equivalent DFA

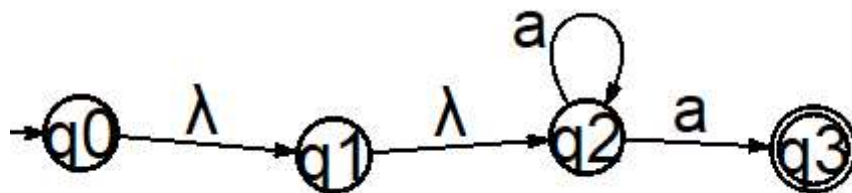
[They accept different languages.](#) [The NFA accepts a superset of the language accepted by the DFA.](#) [They accept the same language.](#) [The DFA accepts a superset of the language accepted by the NFA.](#) (0.5)

- 2) . The minimum number of states required in a DFA to accept a language $L = \{a^n b^{2m} \mid n, m \geq 1\}$

(0.5)

 n m 4 5

- 3) What is the minimum length of string to reach the final state in the below given NFA.



(0.5)

 0 1 2 4

- 4) If $w = \text{abbab}$ the the set of all prefixes of the string w is given as:

[{ a, ab, abb, abba, abbab }.](#) [{ a, ab, abb, abba, abbabbb }.](#) [{ lambda, a, ab, abb, abba, abbab }.](#) [{ lambda, a, ab, bb, abb, bba, abba, abbab }.](#) (0.5)

- 5) Find the regular expression for all strings of size at least two over $\Sigma = \{0, 1\}$ (0.5)

$(0+1)(0+1)(0+1)^*$ $(0+1)^*(0+1)^*$ $(0+1)^*(0+1)$ $(0+1)(0+1)$

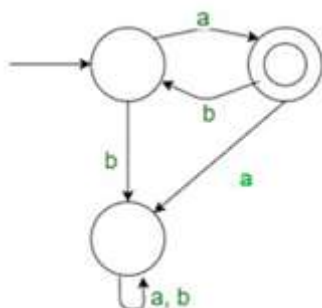
- 6) Identify the non-regular grammar from the following

$S \rightarrow abS \mid a$ $S \rightarrow S1ab, S1 \rightarrow S1ab \mid S2, S2 \rightarrow a \mid \lambda$ $S \rightarrow abS, S \rightarrow bB \mid \lambda, B \rightarrow b$ $S \rightarrow aAB \mid a, A \rightarrow Ba \mid b, B \rightarrow b$ (0.5)

- 7) Which of these will match the strings Rover, rover, and their plural variants only?

$(r+R)over(s)^*$ $(R)rover(s)$ $(r.R)over(s)^*$ $(r+R)over(s+\lambda)$ (0.5)

- 8) Find the regular expression for the given DFA.



aba^* $(ab)^*a$ abb^* aba (0.5)

- 9) How many strings of length less than 4 contain the language described by the regular expression $(x+y)^*y(a+ab)^*$?

7 10 11 12 (0.5)

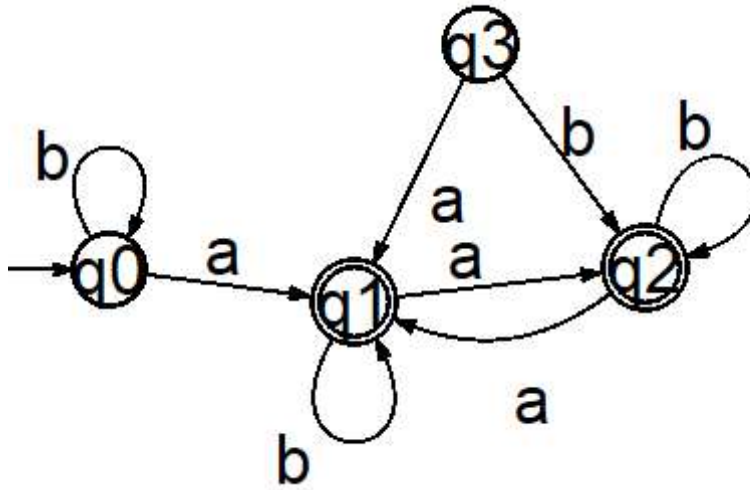
- 10) Find the equivalent of the Regular Expression with $A = a^*$, $B = b^*$ for $AB = ?$

$\{a^n b^n \mid n > 0\}$ $\{a^m b^n \mid m > 0, n > 0\}$ $(a+b)^*$ $(ab)^*$ (0.5)

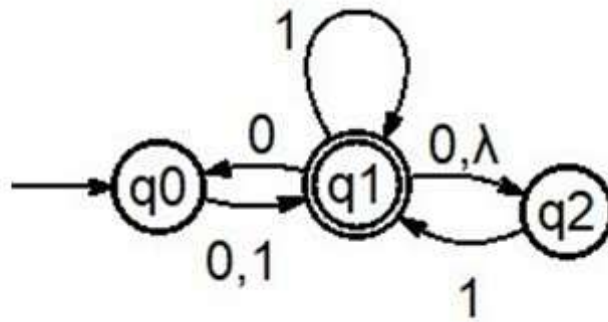
DESCRIPTIVE

Answer all the questions.

- 11) Minimize the states of the DFA given below using mark and reduce method (4)



- 12) Using the pumping lemma prove that the language $L = \{0^k 1^k : k \geq 0\}$ is not regular. (3)
- 13) Using GTG method, find the regular expression for the language $L = \{w \in \{a, b\}^* : n_a(w) \text{ is odd and } n_b(w) \text{ is even}\}$. (3)
- 14) Find L_1/L_2 by constructing a DFA for the language $L_1 = L(a^*baa^*)$ and modify the DFA to accept L_1/L_2 for the language $L_2 = L(aba^*)$. (3)
- 15) Convert the NFA given above to its equivalent DFA (3)



- 16) Find the grammar for the Language $L = \{w : |w| \bmod 3 > 0\}$ on $\Sigma = \{a\}$. (3)
- 17) Give the right linear grammar to generate a language $L = \{00^*(01+0)^*\}$ on $\Sigma = \{0,1\}$. (2)
- 18) Construct a regular expression for the set of all strings over $\{0, 1\}$ that do not contain the substring "101". (2)

- 19) Construct an NFA with three states to accept the language $L=(ab,abc)^*$ on $\Sigma=\{a,b,c\}$. (2)